Code No: 155CX **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
PRINCIPLES OF PROGRAMMING LANGUAGES
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define syntax and semantics.	[2]
b)	What is derivation and a parse tree? Write their significance with examples.	[3]
c)	What are the problems with case sensitive names? Explain.	[2]
d)	What is the scope and life time of a variable? Explain.	[3]
e)	What is a Co-routine? Explain.	[2]
f)	What is generic subprogram?	[3]
g)	Explain about message passing.	[2]
h)	What advantages do monitors have over semaphores? Discuss.	[3]
i)	What are the data types supported in Python?	[2]
j)	Differentiate between procedural and data abstraction.	[3]

PART – B

(50 Marks)

2.a) What are the three general methods of implementing a programming language?
b) Explain with an example how operator associativity can be incorporated in grammars. [5+5]

OR

3.a) The levels of acceptance of any language depend on the language description. Comment on this.
b) In what fundamental way do operational semantics and denotational semantics differ? [5+5]

4.a) What are the design issues for character string types? Discuss.
b) Describe the process of Array initialization. [5+5]

OR

5.a) What are the problems associated with Unconditional Branching? Explain.
b) Explain the differences between subtypes and derived types. [5+5]

6.a) What are the design issues that are involved in functions? Discuss.
b) Explain how subprogram names are passed as parameters. Illustrate with examples. [5+5]

OR

7.a) What are the advantages and disadvantages of dynamic type binding? Discuss.
b) Explain how subprogram is overloaded? Give examples. [5+5]

8.a) Discuss about concurrency in Functional languages.
b) What is a semaphore? What are the operations on semaphores? [5+5]

OR

9.a) Explain in detail about Exception handling in Ada.
b) Write a detailed note on C# threads. [5+5]

10.a) What are the differences between Imperative and functional languages? Explain.
b) Explain the Basic primitives of LISP. Give suitable examples. [5+5]

OR

11.a) What are the applications of functional languages? Explain.
b) Write a detail note on Scripting languages. [5+5]

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